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END SEMESTER EXAMINATION - 2025

Name of the Course: B. Tech. Mech.
Name of the Paper: Mechanical Vibrations

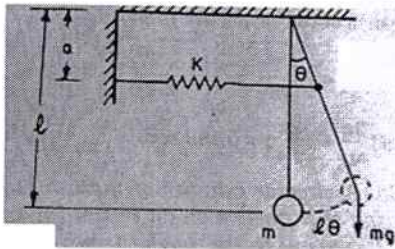
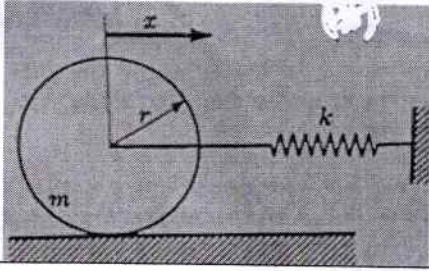
Semester: 6th
Paper Code: TME 610

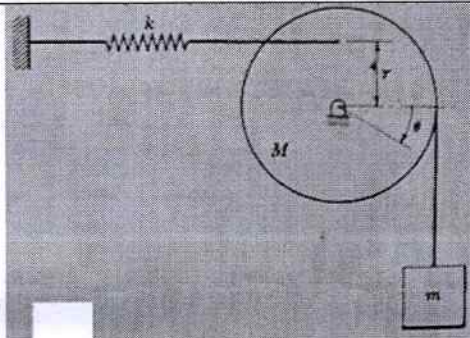
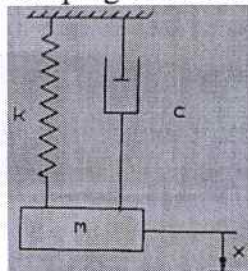
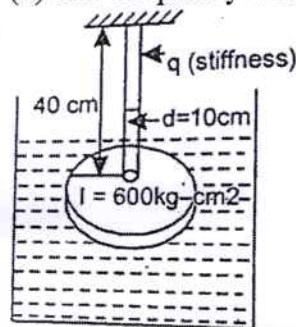
Time: 3 Hours

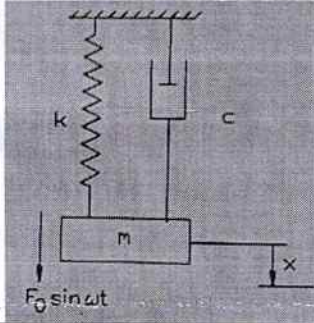
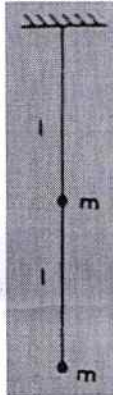
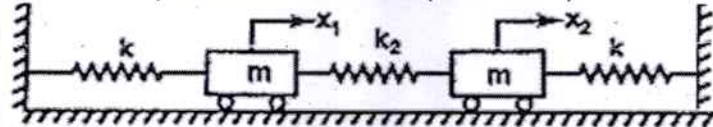
Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks.

Q1	(20marks)	
(a)	Define the following terms related to vibrations i. Cycle ii. Period iii. Frequency iv. Degree of freedom v. Simple Harmonic Motion	CO1
(b)	Add the following two motions analytically and check the solution graphically. $x_1 = 4\cos(\omega t + 10^\circ)$, $x_2 = 6 \sin(\omega t + 60^\circ)$	
(c)	i. Express the complex number $4+j7$ in exponential form. ii. Express the exponential number $5e^{j0.2}$ in complex form.	
Q2	(20 marks)	
(a)	Determine the equations of the motions and natural frequency of the Simple pendulum mass m , length l , spring of stiffness k attached at distance a , from hanging point. 	CO2
(b)	Determine the equations of the motions and natural frequency of the system shown in Figure. 	
(c)	Determine the equations of the motions and natural frequency of the pulley with mass M , mass moment of inertia I , radius R , attached with mass m at radius R and spring of stiffness k at radius r .	

		
Q3	(20 marks)	
(a)	i. Define damping ratio and critical damping with mathematical expressions. (5 Marks) ii. Define Logarithmic decrement with a mathematical expression. (5 Marks)	CO3
(b)	Derive the expression of response of spring-mass-damper (free vibrations) of an underdamped system. (mass=m, spring stiffness=k, damping coefficient=c) 	
(c)	Fig. shows a torsional pendulum with a disc of moment of inertia $I = 600 \text{ kg cm}^2$. The pendulum is immersed in a viscous liquid. During vibrations of pendulum, the successive amplitude decays with 66 % of initial value. Take $G = 4.4 \times 10^{10} \text{ N/m}^2$ for shaft. Determine (a) logarithmic decrement (b) damping torque per unit velocity (c) the periodic time of vibration (d) The frequency when the disc is removed from the liquid. 	
Q4	(20 marks)	
(a)	i. Define Force transmissibility with a mathematical expression. ii. Describe frahms reed tachometer.	CO4
(b)	A 1000 kg machine is mounted on four identical springs of total spring constant k and having negligible damping. The machine is subjected to a harmonic external force of amplitude $F_0 = 490 \text{ N}$ and frequency 180 r.p.m. Determine the amplitude of motion of the machine and maximum force transmitted to the foundation because of the unbalanced force when $k = 1.96 \times 10^6 \text{ N / m}$	

(c)	Derive the expression of the magnification factor and phase difference of spring-mass-damper forced vibration system. (mass=m, spring stiffness=k, damping coefficient=c, harmonic excitation $F = F_0 \sin \omega t$) 	
Q5	(20 marks)	
(a)	Find the frequencies and mode shapes of the system shown in Fig. (Double pendulum) 	CO5 CO6
(b)	Find the frequencies and mode shapes of the system shown in Fig.. 	
(c)	Find the frequencies and mode shapes of the system shown in Fig. (two masses fixed on a tightly stretched string). 